

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 6-7

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1. TITLE

NPM and package.json in Node.js

2. AIM

To understand NPM and **package.json**, install an external package, and use the package in a Node.js application.

3. OBJECTIVE

The objectives of this experiment are:

- To understand NPM and its role in Node.js projects.
- To understand the purpose of the **package.json** file.
- To learn how dependencies are installed and managed.
- To install and use the external **lodash** package.
- To execute and verify a Node.js application using the installed package.

4. THEORY

NPM (Node Package Manager) is the package manager used with Node.js to install, update, and manage JavaScript packages and project dependencies.

package.json is the configuration file of a Node.js project. It stores project information such as the project name, version, main file, scripts, and dependencies.

External package: For this experiment, **Lodash** is installed as an external dependency. Lodash provides utility functions for common JavaScript operations.

Basic NPM workflow:

Create Node.js project
↓
npm init -y
↓
npm install lodash
↓
Import lodash in app.js
↓
Use lodash functions and run the application

5. ALGORITHM / PROCEDURE

1. Create a new Node.js project folder.
 2. Open the project folder in the terminal.
 3. Initialize the project using **npm init -y**.
 4. Install the external Lodash package using **npm install lodash**.
 5. Verify that Lodash is added to the project dependencies.
 6. Create **app.js** and import the Lodash package.
 7. Create an array of numbers.
 8. Use Lodash functions to calculate the sum, maximum, and minimum values.
 9. Run the application using **node app.js**.
 10. Verify the actual output in the terminal.
-

Important NPM commands

```
npm init -y
npm install lodash
npm install
node app.js
```

6. PROGRAM

app.js

```
// Import the external Lodash package
const _ = require("lodash");

const numbers = [10, 20, 30, 40, 50];

// Calculate sum using Lodash
const sum = _.sum(numbers);

// Find the maximum value
const maximum = _.max(numbers);

// Find the minimum value
const minimum = _.min(numbers);

console.log("Original Array:", numbers);
console.log("Sum:", sum);
console.log("Maximum:", maximum);
console.log("Minimum:", minimum);
```

Expected project structure

```
Q7-NPM-and-package-json/
├──
├── app.js
├── package.json
├── package-lock.json
```

The **node_modules** directory is generated by NPM after installation and normally should not be uploaded to GitHub.

7. CODE EXPLANATION

Code / Method	Explanation
<code>const _ = require("lodash")</code>	Imports the installed Lodash package into the Node.js application.
<code>const numbers = [...]</code>	Creates the array used for the demonstration.
<code>_.sum(numbers)</code>	Calculates the sum of all values in the array.
<code>_.max(numbers)</code>	Finds the largest value in the array.
<code>_.min(numbers)</code>	Finds the smallest value in the array.
<code>console.log()</code>	Displays the results in the terminal.

8. TEST CASES AND EXECUTION DATA

The NPM commands and Node.js program were executed successfully. The actual terminal output was used for verification.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Run npm init -y	package.json should be created.	package.json created successfully.	Pass
TC-02	Run npm install lodash	Lodash should be installed.	1 package added; 0 vulnerabilities found.	Pass
TC-03	Run node app.js	Array and calculated values should be displayed.	Array, Sum 150, Maximum 50, Minimum 10 displayed.	Pass

9. OUTPUT

Output 1: NPM Project Initialization, Package Installation and Program Execution

```
c:\Users\dadhi\FULL_STACK\01-observation-and-record\observation\Task 6\Q7-NPM-and-package-json\src>npm init -y
Wrote to c:\Users\dadhi\FULL_STACK\01-observation-and-record\observation\Task 6\Q7-NPM-and-package-json\src\package.json:

{
  "name": "src",
  "version": "1.0.0",
  "description": "",
  "main": "app.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

c:\Users\dadhi\FULL_STACK\01-observation-and-record\observation\Task 6\Q7-NPM-and-package-json\src>npm install lodash

added 1 package, and audited 2 packages in 3s

found 0 vulnerabilities

c:\Users\dadhi\FULL_STACK\01-observation-and-record\observation\Task 6\Q7-NPM-and-package-json\src>node app.js
Original Array: [ 10, 20, 30, 40, 50 ]
Sum: 150
Maximum: 50
Minimum: 10

c:\Users\dadhi\FULL_STACK\01-observation-and-record\observation\Task 6\Q7-NPM-and-package-json\src>
```

9. OUTPUT

Observed execution results

Item	Observed Value
Project initialization	package.json created successfully.
External package	lodash installed successfully.
Installation result	1 package added; 2 packages audited; 0 vulnerabilities found.
Original Array	[10, 20, 30, 40, 50]
Sum	150
Maximum	50
Minimum	10

10. RESULT

Thus, NPM and **package.json** were successfully demonstrated. The Lodash external package was installed and used in a Node.js application to calculate the sum, maximum, and minimum of an array. The application was executed successfully and the output was verified using the actual terminal execution.